MANCHESTER 1824
The University of Manchester

Understanding the Earth

Workshop 10 Is our solar system "average"?

Professor Jamie Gilmour (Professor of Planetary Science)

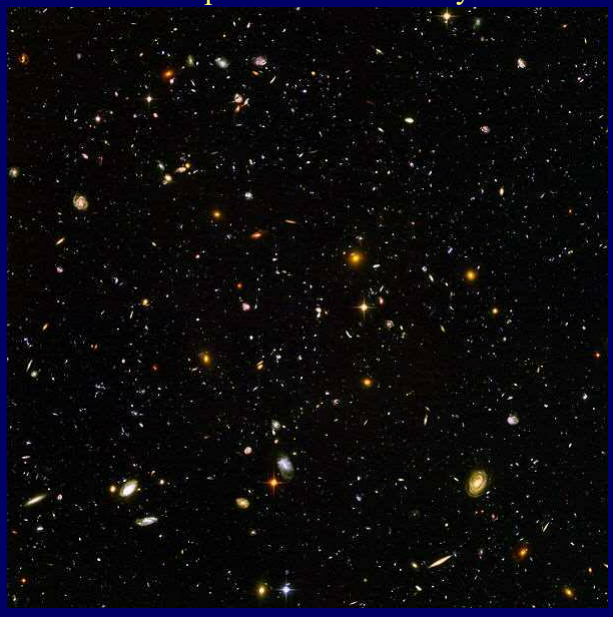
Planetary Science

- Contrasts…
  - History & properties of our planet/solar system
  - General history & properties of planets and stellar systems.
- Asks "Is ours typical or unusual?"

Today: Is our solar system "average"?

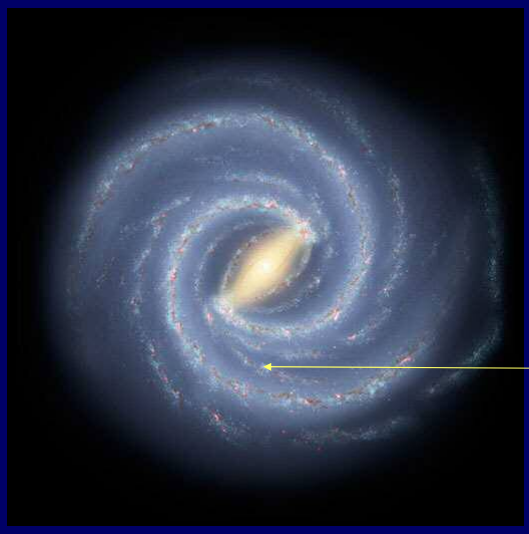
Where are we?
When are we?

~10^4 Galaxies in "~1 part in 10^7 of the sky".



http://en.wikipedia.org/wiki/Hubble_Ultra_Deep_Field

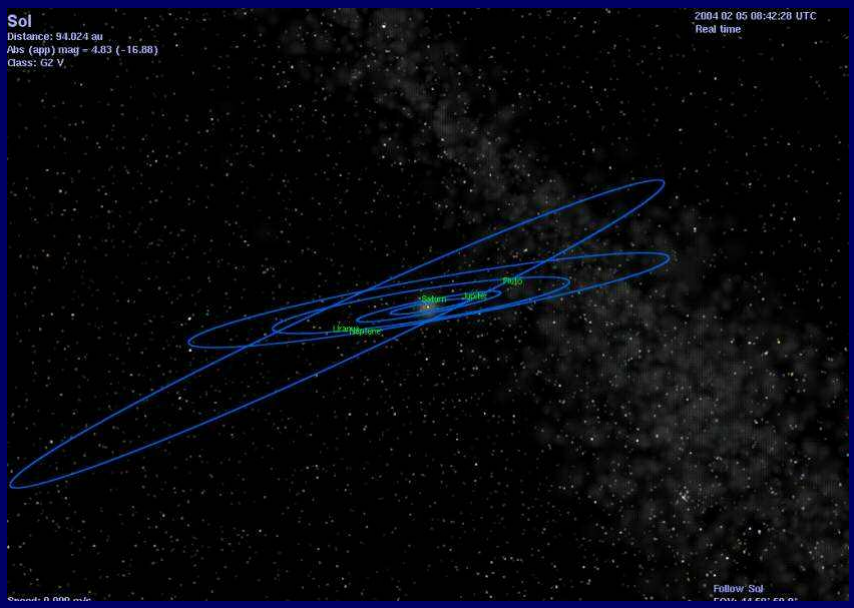
Milky Way Galaxy 200-400 billion stars



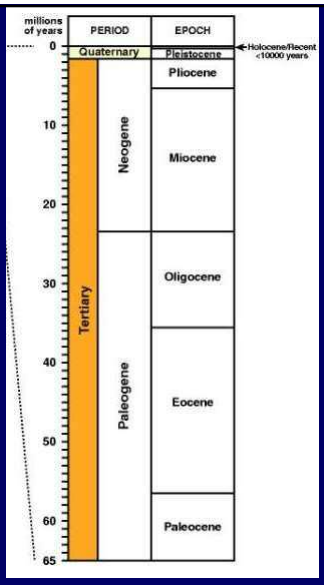
You are here (approximately)

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http://www.nasa.gov/mission_pages/spitzer/multimedia/20080603a.html

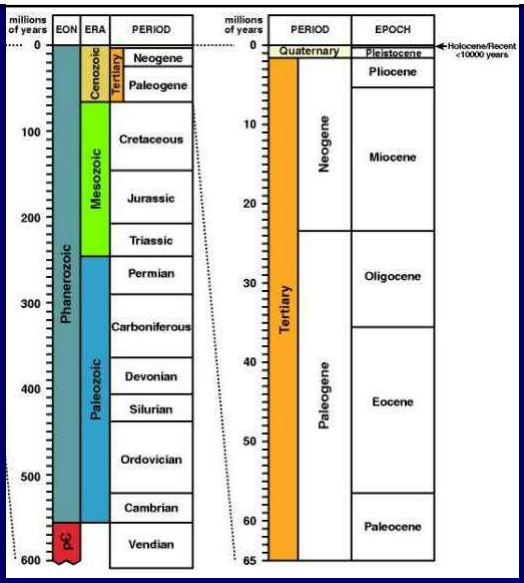


Time Scales



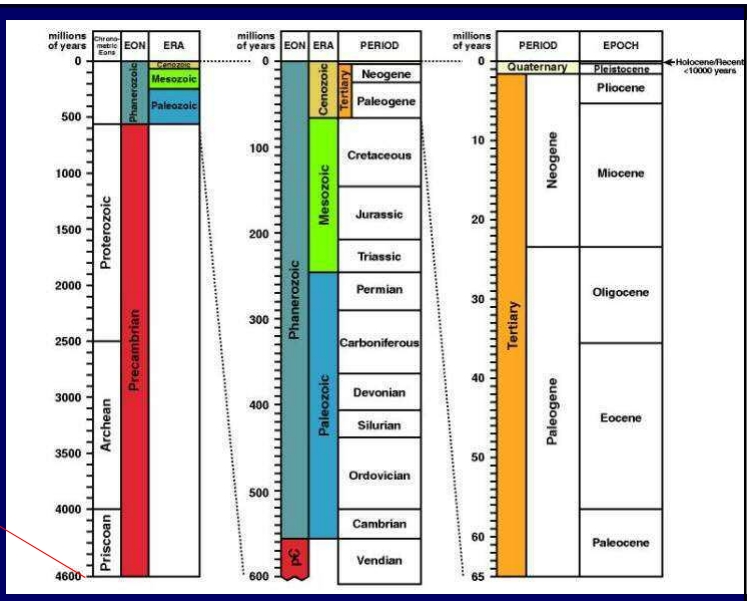
http://www.talkorigins.org/faqs/timescale.html

Time Scales



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Time Scales

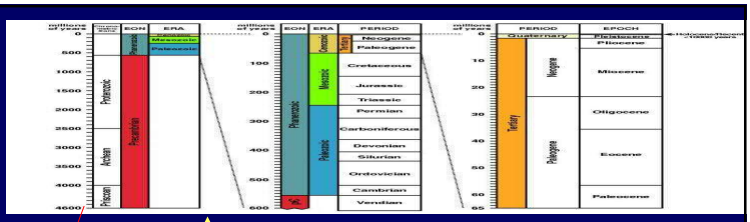


Age of the solar system: ~4.567 billion years, ~150 Ps

(Peta = 10^{15})

http://www.talkorigins.org/faqs/timescale.html

Time Scales



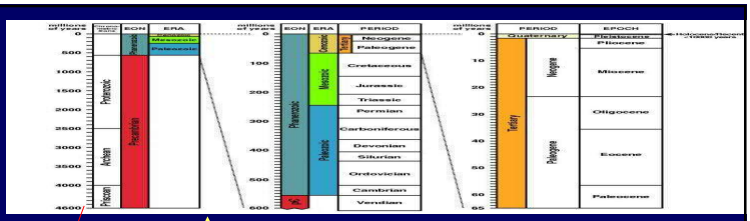
Age of Earth and solar system: 4.5 Billion years

Approx 9 billion years (twice age of Earth) of history before Earth forms. (300 Ps)

Origin of Universe: ~13.8 Billion years. (440 Ps)

http://www.talkorigins.org/faqs/timescale.html

Time Scales



Age of Earth and solar system: 4.5 Billion years

Generations of stars make the heavier elements that are present in our solar system.

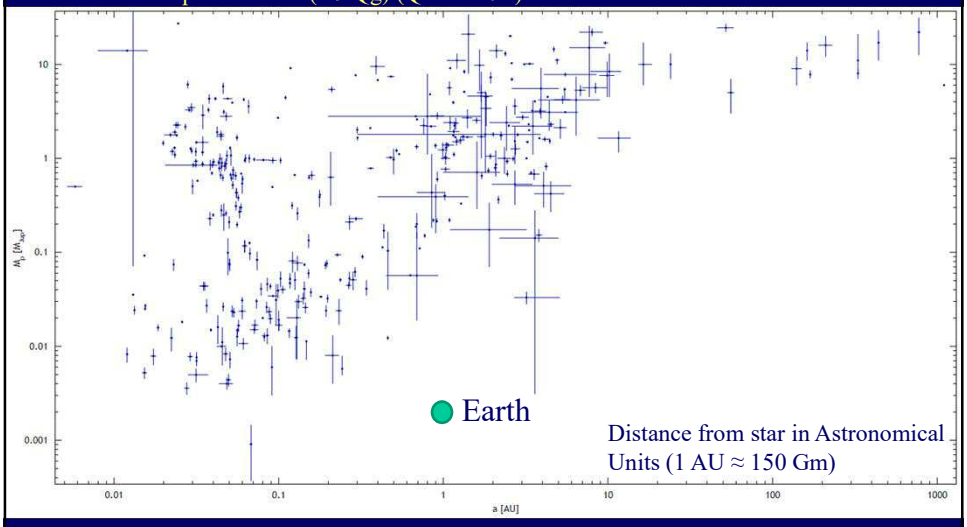
Big bang makes H and He.

Origin of Universe: ~13.8 Billion years.

http://www.talkorigins.org/faqs/timescale.html

Exoplanets.

Planet Mass in Jupiter Masses (1.9 Qg) (Quetta 10^{30})



https://dace.unige.ch/dashboard/index.html

What do you notice, about this graph. Can you explain any features?

Lots of time, lots of space…

So where are the alien civilizations?

What things do all these aliens have in common?
(How many films/franchises do you recognise? Do you have a favourite?)












Lots of time, lots of space…

So where are the alien civilizations?

Ways to think about this…

Fermi's Argument

If they existed, they would be here - *Enrico Fermi*

- Galaxy scale is ~100,000 light years* (90 Pm).
- In a few centuries we might expect to be able to
  - build a probe capable of 0.01c**.
  - make the probe self-replicating on arrival at another system. A ***Von Neumann probe***. (Could be dangerous… Universal Paperclips**)
  - Takes 10 million years to colonise the galaxy
- The galaxy is ~13.8 billion years old

*1 light year is the distance light travels in one year.
**c is the speed of light in a vacuum.

**https://www.decisionproblem.com/paperclips/ (if you have a few days to waste)

Where are the alien civilizations?

Should we expect to have evidence of them?
Fermi says “Yes”.

But do they exist?

Several decades ago a way of addressing the question was proposed, now known as... “The Drake Equation”.

Using it relies on the *Principle of Mediocrity*.

Our ideas about the universe.

- Geocentric view – the Earth was the centre of the universe. Special
- Heliocentric view – placed the sun at the centre of the universe.
- Modern view – the sun is just one star in a galaxy, and there are many galaxies.

Principle of Mediocrity

There is nothing special about the solar system or the planet Earth.

We view the universe from an average position, as if we picked a place and time at random.

How the argument works…

- Assume our solar system is typical of planetary systems – assume the principle of mediocrity is correct.
- Use what we know about it (and can see from it) to estimate how many other civilizations there are in the galaxy right now
  - The Drake Equation.
- If the estimate suggests there are lots of civilizations out there, the fact that we haven't been visited suggests our assumption is wrong.
- This would our solar system has some unusual features that make it favourable for civilizations like ours.
- What might they be?

The Drake Equation

$$N_C = N^* f_p n_e f_l f_i f_c f_L$$

- N_c = Number of communicating/travelling civilizations in the galaxy today.
- N^* = Number of stars in our galaxy.
- f_p = Fraction of stars with planets.
- n_e = Planets per star capable of supporting life.
- f_l = fraction of n_e where life starts.
- f_i = given life, fraction intelligence evolves.
- f_c = fraction of intelligent life that develops technology to enable space exploration (Fermi style).
- f_L = fraction of planet's lifetime for which exploration continues.

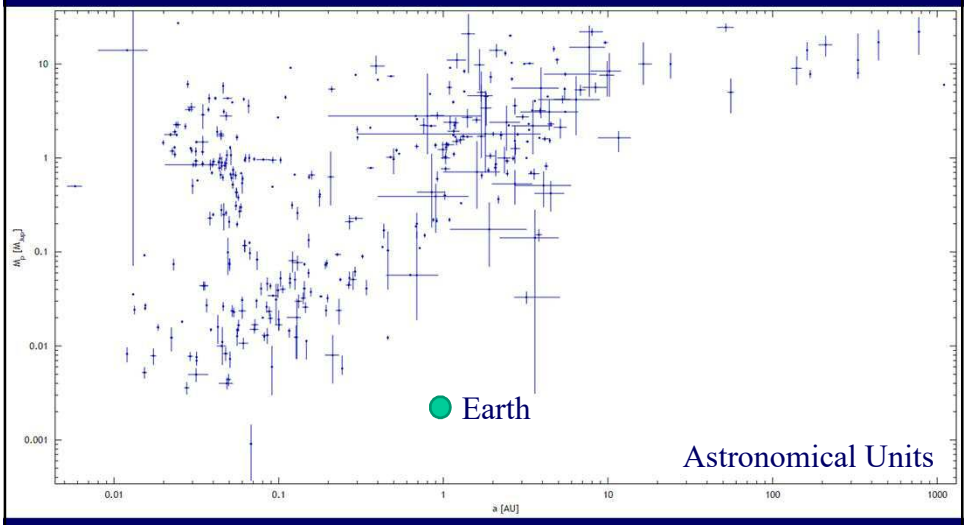
The Drake Equation

$$N_C = N^* f_p n_e f_l f_i f_c f_L$$

- N^* = Number of stars in our galaxy. 2-4 x 10^{11}.
- What fraction *f* have planets?

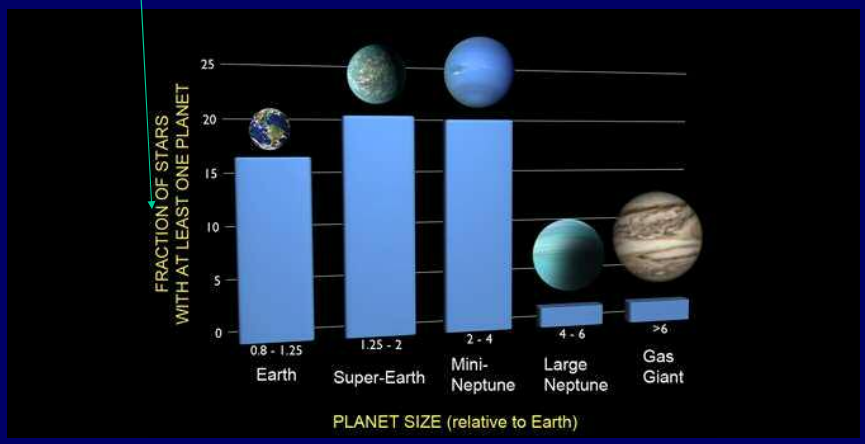
Exoplanets.

Jupiter Masses



https://dace.unige.ch/dashboard/index.html

(Percent)



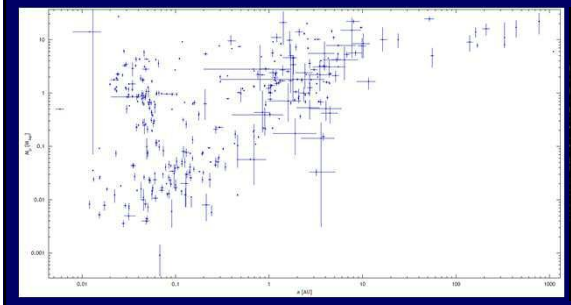
https://www.nasa.gov/mission_pages/kepler/news/17-percent-of-stars-have-earth-size-planets.html

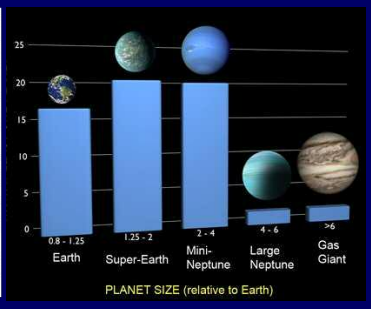
The Drake Equation

$$N_C = N^* f_p n_e f_l f_i f_c f_L$$

- N* = Number of stars in our galaxy. 2-4 x 10^{11}.
- f_p = Fraction of stars with planets. 100%
- n_e = Planets per star capable of supporting life.

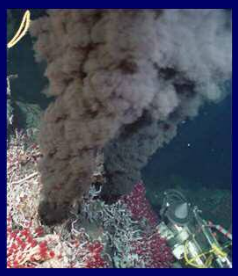
Estimate the number of planets per star capable of supporting life.





- What is life?
- What does it need?
- What fraction of planets in our solar system might support life?

- What does life need?
- (Think in as general terms as possible.)



Hydrothermal vent



Noodle



Tree

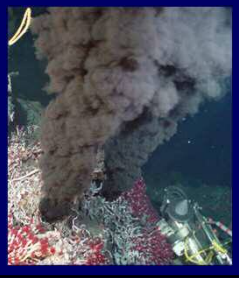
"A venting black smoker emits jets of particle-laden fluids. […]

"Underwater volcanoes at spreading ridges and convergent plate boundaries produce hot springs known as hydrothermal vents. Scientists first discovered hydrothermal vents in 1977 while exploring an oceanic spreading ridge near the Galapagos Islands. To their amazement, the scientists also found that the hydrothermal vents were surrounded by large numbers of organisms that had never been seen before. These biological communities depend upon chemical processes that result from the interaction of seawater and hot magma associated with underwater volcanoes.

https://oceanservice.noaa.gov/facts/vents.html

- What does life need?
  - A source of useful** energy ("free energy")… disequilibrium.
  - What sort of disequilibrium is exploited by…
  1. A microbial community round a black smoker
  2. An animal at the surface of the Earth.
  3. A green plant at the surface of the Earth.

** We can use energy if it moves form a part of a system with a high energy density to a part with a low energy density (concept of *entropy).* The presence of high and low energy density regions is what we mean by *disequilibrium*. Example – hot and cold regions.







Where is there disequilibrium in the solar system?

All over the place.

Where is there disequilibrium in the solar system?

Can we narrow the requirements down?

We look for “life like ours”

Where is there disequilibrium in the solar system?

Can we narrow the requirements down?

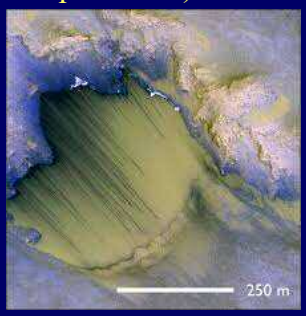
We look for "life like ours"

- Carbon based
- Uses water as a solvent.

Hence "follow the water".

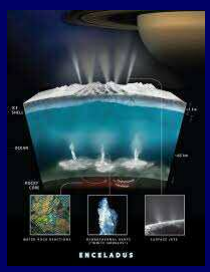
- What fraction of planets in our solar system might support life?

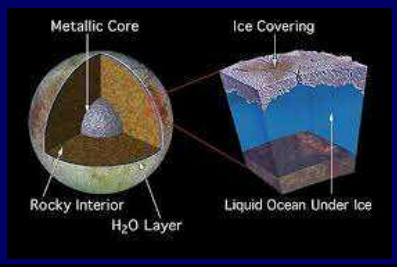
Mars (Recurring Slope Lineae)



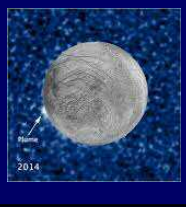


Enceladus





Europa



The Drake Equation

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- Nc = Number of communicating civilizations in the galaxy today.
- N* = Number of stars in our galaxy.
- fp = Fraction of stars with planets.
- ne = Planets per star capable of supporting life.
- f_l = fraction of n_e where life starts.
  - What's your estimate of f_l?
  - What do you know that is relevant?
  - How might we get a better estimate?

The Drake Equation

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- f_i = given life, fraction where intelligence evolves.
- f_c = fraction of intelligent life that develops technology to enable space exploration.
- *What does a planet need to enable these two steps?*

The Drake Equation

$$N_C = N^* f_p n_e f_l f_i f_c f_L$$

- N^* = Number of stars in our galaxy. 2-4 x 10^8.
- f_p = Fraction of stars with planets. ~100%
- *n_e = Planets per star capable of supporting life. ~2*
- *f_l = fraction of n_e where life starts. ~1*
- *f_i = given life, fraction where intelligence evolves. (50%)*
- *f_c = fraction of intelligent life that develops technology to enable space exploration. (50%)*
- *f_L = fraction of planet's lifetime for which travel continues.~ 10^{-6} (~1000 years)*

Number of communicating civilisations in the galaxy is…

150,000 (my answer).

So where are they? We have not seen a *Von Neumann* probe.

The Drake Equation

$$N_C = N^* f_p n_e f_l f_i f_c f_L$$

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Why might our solar system be special?

- It supports a technological civilization.
- A civilization that can compare its environment with other places in the universe.
- How common are these (or places where they might form and survive)?

- If they are rare, then something about our solar system might be special.
- The absence of alien visitors tests the Principle of Mediocrity

Resolutions?

- The Drake equation suggests there are and have been many other civilizations out there in the galaxy.
- Fermi's argument suggests there haven't been any.

The Drake Equation

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- f_L = *fraction of planet's lifetime for which travel continues.~ 10^{-6}*

If there are multiple hurdles between life and intelligence, the most likely situation observed by intelligence is an almost immediate start. This might just mean there are multiple hurdles and intelligence is very rare.

"Travel" is in our future. Perhaps many species get to a state like ours, then never make it to the "travel" level. A "Doomsday argument".

The zoo hypothesis.

Estivation.

Add your idea here…